

FaceBook :

Functional requirements:

user should be able to post images based on his preference,
user should be able to search images among the existing images,
user's feed should get updated after every 5seconds

Non-functional requirements :

data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:

user,
feed,
post,
history

API's :

POST:/posts/upload

```
{  
  userID,  
  {filemetadata}  
}
```

GET:/search/{postID}

HLD :

